Name: ______________________________ Tutorial group: TX

Matriculation number: | | | | | | | | | |

17 September 2019 **MH2500 Test 1** 40 minutes

QUESTION 1. **(10 marks)**

Suppose that the cumulative distribution function (CDF) of X is given by

$$F(x) = \begin{cases} 0, & x < 1; \\ 1/3, & 1 \le x < 2; \\ 5/6, & 2 \le x < 4; \\ 1, & x \ge 4. \end{cases}$$

(a) (2 marks) Find $\Pr\{1 < X < 2\}$ and $\Pr\{2 \le X < 3\}$. (No partial credits will be given for each probability.)

(b) (8 marks) Compare $\log_2 \mathbb{E}(X)$ with $\mathbb{E}(\log_2 X)$. Which is larger? State your reason.

QUESTION 2. **(8 marks)**

Let events A and B be independent, with $\Pr(A) = p$ and $\Pr(B) = 0.3$. Let C denote the event that at least one of the events A and B occurs and D denote the event that exactly one of the events A and B occurs.

(a) (2 marks) Find $\Pr(C)$. Express your answer in terms of p.

(b) (3 marks) Find $\Pr(D)$. Express your answer in terms of p.

(c) (3 marks) Can C and D be independent? If this is possible, find all possible values of p such that C and D are independent; otherwise state your reason why C and D can never be independent.

QUESTION 3. **(11 marks)**

Customers arrive at a shop randomly with a rate of 4 per hour, modelled by a Poisson process. In a one-hour period, calculate the conditional probability that no customers arrive in the second half hour given that

(a) (4 marks) exactly 4 customers arrive in the first half hour;

(b) (7 marks) exactly 4 customers arrive over the whole one-hour period.